

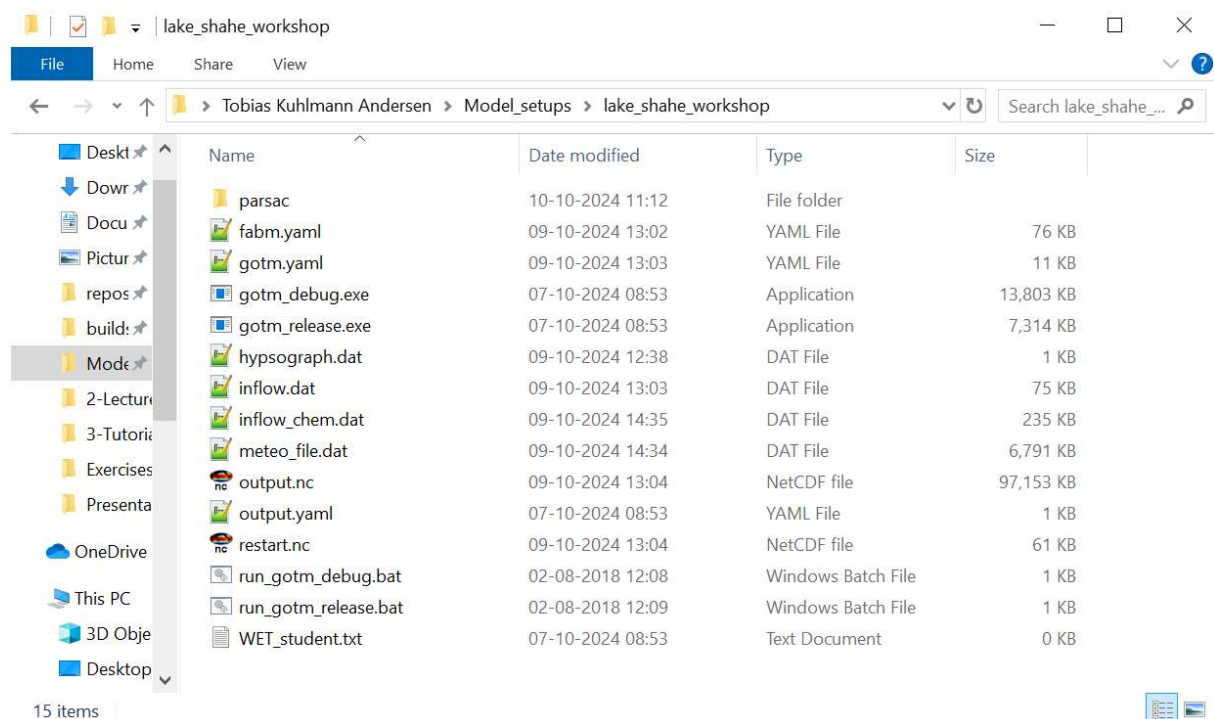
## 3-day workshop on modelling lake ecosystems with WET: Tutorial on auto-calibration with parsac

This tutorial introduces installation and application of the parsac sensitivity analysis and auto-calibration tool by Jorn Bruggeman of Bolding and Bruggeman ApS to a workable GOTM-WET model setup. On the use of parsac, this tutorial should be used as a supplement to the information provided on <https://pypi.org/project/parsac/> and <https://bolding-bruggeman.com/portfolio/parsac/> (please visit these sites for more information). For calibration, parsac applies the parallel direct search method Differential Evolution (Storn and Price 1997) to estimate the most optimal choice of model parameter values within predefined, parameter-specific ranges based on optimization of a Maximum Likelihood multi-objective function.

This tutorial builds upon the lecture on calibration and model evaluation of complex models and aims to teach how to perform supervised auto-calibration of your GOTM-WET model with parsac. The tutorial requires a workable GOTM-WET model set-up, observations to be used in the calibration and installed Python package parsac.

### Configuring your lake model directory for parsac use

You will need a working lake model directory before you can start auto-calibration. In practice, users can first create a complete QWET model project, then exit the QWET (and QGIS) interface, and then navigate to the model: \\WET\_model\_example\\project\_name\\LAKEMODEL\\Default. Make a copy of the model QWET default directory into a new folder (preferably on your local drive). Or users can configure the model directory by themselves. Below is an example of a working model directory for Shahe Reservoir:



**Required materials****Workable GOTM-WET setup model directory****Observation files**

In the bottom of the parsac xml configuration file, users must point to the file(s) containing observations, which will be used to train the model through the parameter calibration. The format of the observation files matches exactly that of the observation files that may imported into QWET.

Observations for Shahe Reservoir are provided, so you do not need to prepare obs files for Shahe case study.

*Preparing .obs files for QWET or parsac*

If you work with another case study, here is how to prepare .obs files. The observation files must comprise four columns including (in this order):

1. Date [yyyy-mm-dd]
2. Time [hh:mm:ss]
3. Depth from surface [m] (Negative values. For example, -1 represents 1 meters depth below surface)
4. Value of state variable concentration (in units corresponding to model output units, e.g. Celsius for temperature, mg/L for nutrients and oxygen, and µg/L for chlorophyll a)

An example of an \*.obs file of correct format can be found [here](#). The names of the individual observations files must match the working .xml file in the “parsac” folder. For further details, please watch this [tutorial video](#) on how to format and work with observations in QWET.

**parsac**

parsac is a Python-based tool for sensitivity analysis and auto-calibration in parallel developed by Bolding&Bruggeman. It is designed for analysis of models that take significant time to run. To download parsac, you will need a Python working environment and can install via Python pip: “pip install parsac –user”. Parsac requires the Python package Parallel Python to parallelize the auto-calibration process. For more details on parsac, we recommend checking out its Github page: <https://github.com/BoldingBruggeman/parsac>.

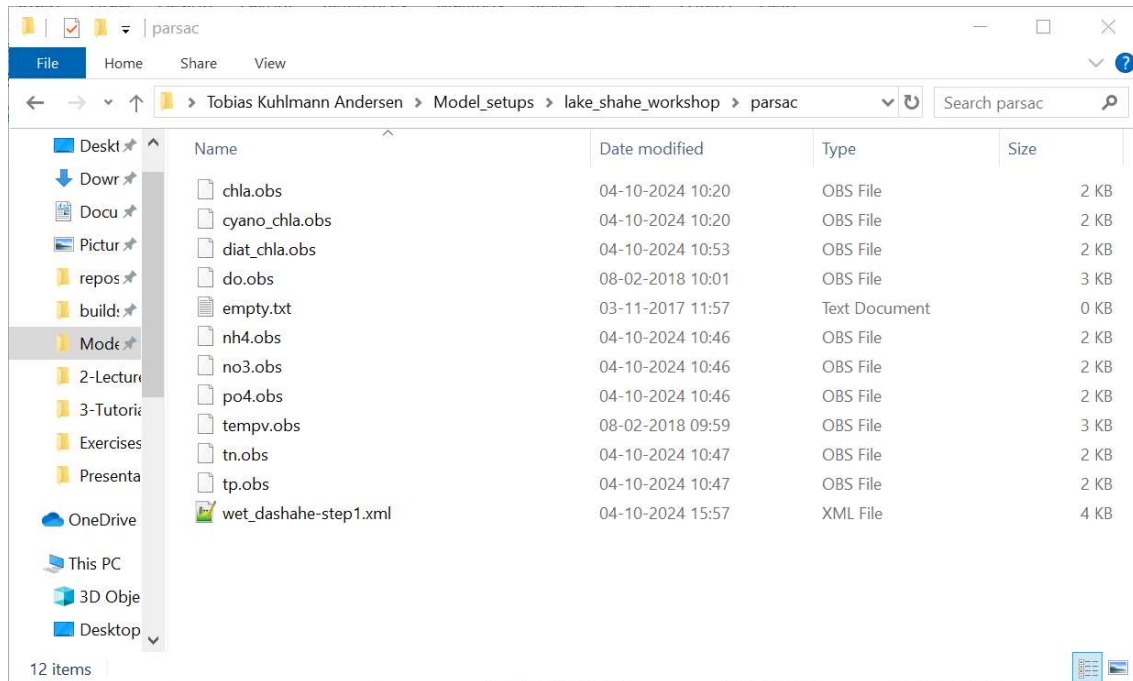
For installing Python 3, parsac, and required packages (pp (parallel python) and SALib), see installation guide “Tutorial on parsac and Python 3 installer” provided on USB stick Day 2 or in WeChat group.

**DB Browser for SQLite (optional)**

If you want to open the database file parsac creates with model simulation information, you can download the [latest version DB Browser for SQLite](#) or use the Windows installer “DB.Browser.for.SQLite-v3.13.0-win64.msi” in *Materials folder* provided on Day 1 by USB stick.

**Folder setup for parsac**

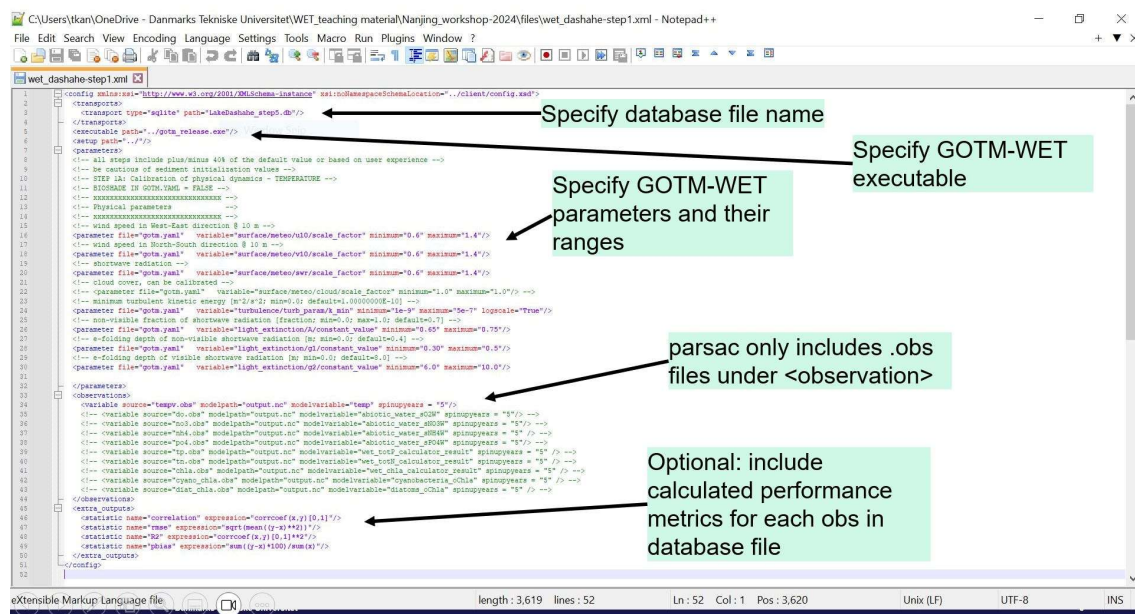
Within your working model folder, you will need to create a folder named *parsac*. In the *parsac* folder you should include observation files (\*.obs) for lake and a xml file with the configurations of parsac. Optionally, you can include bat files to execute the different parsac steps (see below). Below is an example of a parsac folder for the Shahe Reservoir:



### Configuring parsac xml file

All calibration steps with parsac are configured through an xml file, which allows users to point to the parameters targeted for calibration and assign their ranges (and add or remove parameters for calibration). Be aware that parsac or GOTM-WET does not check if parameter values are within theoretical and technical ranges. So, users can specify parameter values that are not valid.

A comprehensive example of a xml file (wet\_dashahe-step5.xml) is available in the wet-workshop-Day2\Exercises\Exercise 2 – Calibration.



### Calibration routine with parsac

The calibration routine consists of three actions in parsac: 1) executing the calibration, 2) plotting model performance against parameter value ranges and 3) running best performing model with their respective parsac commands in your Python console (for instance Anaconda Prompt):

1. “parsac calibration run xmlfile”
2. “parsac calibration plot xmlfile”
3. “parsac calibration plotbest xmlfile”

xmlfile refers to the path and xml file name, for instance  
C:/path\_to\_parsac\_folder/config\_parsac.xml

In this tutorial, we refer to a calibration iteration when the modelers have completed the three actions above. Calibration of a GOTM-WET model with parsac will include many calibration iterations and will depend on the number of observations, state variable and parameters included. In this tutorial we also refer to calibration steps, which is a subset of all the calibration iterations that focus on calibrating a specific model process and/or reproduce specific observations (see *Calibration strategy and iterations* below for more information).

### *parsac run*

During calibration (“parsac calibration run”), a database file (\*.db, named as specified in the .xml file) is created to store each model simulation with unique model execution id (id), calibration run (run), selected values for parameters in .xml (parameters), ln-likelihood model performance (lnlikelihood) and optional performance metrics specified in .xml (extra\_outputs). Calibration run is specified as several calibration iterations can be performed on the same xml file to the same database file, which is recorded by parsac.

| id | run | time                | parameters | lnlikelihood      | extra_outputs                            |
|----|-----|---------------------|------------|-------------------|------------------------------------------|
| 1  | 1   | 2024-10-04 05:04:39 | ...        | -281.885586156077 | ("correlation": [0.8708192533175143, ... |
| 2  | 1   | 2024-10-04 05:04:39 | ...        | -461.442231792641 | ("correlation": [0.7970787505675562, ... |
| 3  | 1   | 2024-10-04 05:04:39 | ...        | -1056.13838519606 | ("correlation": [0.9029539530448099, ... |
| 4  | 1   | 2024-10-04 05:04:39 | ...        | -204.044462370799 | ("correlation": [0.976321312262855, ...  |
| 5  | 1   | 2024-10-04 05:04:39 | ...        | -240.927589056718 | ("correlation": [0.9670754344847236, ... |
| 6  | 1   | 2024-10-04 05:04:39 | ...        | -1067.39175187773 | ("correlation": [0.8141064396865688, ... |
| 7  | 1   | 2024-10-04 05:04:39 | ...        | -596.533432228589 | ("correlation": [0.8220177068749519, ... |
| 8  | 1   | 2024-10-04 05:04:39 | ...        | -1781.419741966   | ("correlation": [0.8646100026100927, ... |
| 9  | 1   | 2024-10-04 05:04:39 | ...        | -484.440477872048 | ("correlation": [0.797831169037119, ...  |
| 10 | 1   | 2024-10-04 05:04:39 | ...        | -173.86647142036  | ("correlation": [0.9326689883550299, ... |
| 11 | 1   | 2024-10-04 05:04:39 | ...        | -359.960388974666 | ("correlation": [0.8327777190880113, ... |
| 12 | 1   | 2024-10-04 05:04:39 | ...        | -1031.25883520819 | ("correlation": [0.903077018720387, ...  |
| 13 | 1   | 2024-10-04 05:04:39 | ...        | -375.743587098613 | ("correlation": [0.9025389585669329, ... |
| 14 | 1   | 2024-10-04 05:04:39 | ...        | -368.744214009865 | ("correlation": [0.8783708582275586, ... |
| 15 | 1   | 2024-10-04 05:04:39 | ...        | -233.930079367386 | ("correlation": [0.841208403776316, ...  |
| 16 | 1   | 2024-10-04 05:04:39 | ...        | -42.8246786361463 | ("correlation": [0.976270061213772, ...  |
| 17 | 1   | 2024-10-04 05:04:39 | ...        | -3084.57449324047 | ("correlation": [0.9230737205670546, ... |
| 18 | 1   | 2024-10-04 05:04:39 | ...        | -2339.54150505293 | ("correlation": [0.9724541965792388, ... |
| 19 | 1   | 2024-10-04 05:04:39 | ...        | -267.762957115224 | ("correlation": [0.8270328506919061, ... |
| 20 | 1   | 2024-10-04 05:04:39 | ...        | -128.630933434442 | ("correlation": [0.970002858300136, ...  |
| 21 | 1   | 2024-10-04 05:04:39 | ...        | -3412.39522300707 | ("correlation": [0.9318002241443352, ... |
| 22 | 1   | 2024-10-04 05:04:39 | ...        | -401.799437540449 | ("correlation": [0.845574008556834, ...  |
| 23 | 1   | 2024-10-04 05:04:39 | ...        | -320.112540939835 | ("correlation": [0.9332657462975613, ... |
| 24 | 1   | 2024-10-04 05:04:39 | ...        | -5307.90221989694 | ("correlation": [0.9603215928141937, ... |
| 25 | 1   | 2024-10-04 05:04:39 | ...        | -353.951022190616 | ("correlation": [0.9586284279581805, ... |
| 26 | 1   | 2024-10-04 05:04:39 | ...        | -161.691378891736 | ("correlation": [0.9553476291396821, ... |
| 27 | 1   | 2024-10-04 05:04:39 | ...        | -107.847224399663 | ("correlation": [0.9767269309176041, ... |
| 28 | 1   | 2024-10-04 05:04:39 | ...        | -236.19838953307  | ("correlation": [0.9469632588574199, ... |
| 29 | 1   | 2024-10-04 05:04:39 | ...        | -561.090948351906 | ("correlation": [0.8297110149000436, ... |

### *Weighting of observations with different variables*

If several state variables and their observations are included in the calibration, parsac automatically weights the observations. As Jorn Bruggemann writes on [parsac's wiki page](#):

“parsac’s calibration approach maximizes the (log) likelihood. This combines all model-observations differences (after transformation, if that’s activated), each weighted by the reciprocal of its “standard

deviation" (formally, squares of the differences divided by their variance). This standard deviation is currently a constant for each observed variable, it cannot yet depend on time and/or depth. It can be prescribed in your xml configuration file by adding an attribute `sd="<VALUE>"` to the observed variable. If it is not prescribed, parsac will instead [estimate it from the model-observation differences](#). In that case, a variable that the model cannot capture well will automatically get a higher sd. All observations are assumed to be independent and (thus) uncorrelated."

For details on the consequences of this approach, check out [parsac's wiki page](#).

Notice that users can leverage `sd="<VALUE>"` to manually weigh the optimization and thereby focus the calibration on specified state variables with observations.

### *parsac plotbest*

plotbest function will execute the GOTM-WET model with the best performance in the database file.

In the command window, parsac will print several useful information:

```

C:\windows\system32\cmd.exe
D:\Models\lake_dashahe_qwet\parsac>C:\Users\Administrator\anaconda3\Scripts\parsac calibration plotbest D:\Models\lake_d
ashahe_qwet\parsac\wet_dashahe-step5.xml
Loading cached copy of D:\Models\lake_dashahe_qwet\parsac\tempv.obs...
Loading cached copy of D:\Models\lake_dashahe_qwet\parsac\do.obs...
Loading cached copy of D:\Models\lake_dashahe_qwet\parsac\nh4.obs...
Loading cached copy of D:\Models\lake_dashahe_qwet\parsac\po4.obs...
Loading cached copy of D:\Models\lake_dashahe_qwet\parsac\tp.obs...
Loading cached copy of D:\Models\lake_dashahe_qwet\parsac\tn.obs...
Loading cached copy of D:\Models\lake_dashahe_qwet\parsac\chl_a.obs...
Loading cached copy of D:\Models\lake_dashahe_qwet\parsac\cu_mo_chla.obs...
Loading cached copy of D:\Models\lake_dashahe_qwet\parsac\diat_chla.obs...
Retrieving results...
Found 1029843 results, of which 0 were invalid.
1th best parameter set:
gotm.yaml/surface/meteo/u10/scale_factor = 1.2217
gotm.yaml/surface/meteo/v10/scale_factor = 1.0077
gotm.yaml/surface/meteo/swr/scale_factor = 0.941884
gotm.yaml/turbulence/turb_param/k_min = 3.28413e-06
gotm.yaml/light_extinction/A/constant_value = 0.738787
gotm.yaml/light_extinction/g1/constant_value = 0.398411
gotm.yaml/light_extinction/g2/constant_value = 7.40694
fabm.yaml/instances/abiotic_water/parameters/kDMinPOMW = 0.0629678
fabm.yaml/instances/abiotic_water/parameters/kDMinDOMW = 0.0416152
fabm.yaml/instances/abiotic_water/parameters/cThetaMinPOMW = 1.03687
fabm.yaml/instances/abiotic_water/parameters/cThetaMinDOMW = 1.06267
fabm.yaml/instances/abiotic_sediment/parameters/kDMinPOMS = 0.00998945
fabm.yaml/instances/abiotic_sediment/parameters/kDMinDOMS = 0.00990978
fabm.yaml/instances/abiotic_sediment/parameters/cThetaMinDOMS = 1.05223
fabm.yaml/instances/abiotic_sediment/parameters/cThetaMinPOMS = 1.04707
  
```



```

C:\windows\system32\cmd.exe
fabm.yaml/instances/cladocerans/parameters/cClearPrey2 = 1.54922
fabm.yaml/instances/cladocerans/parameters/cClearPrey3 = 2.03721
fabm.yaml/instances/cladocerans/parameters/cClearPrey4 = 2.48719
fabm.yaml/instances/cladocerans/parameters/fGutOccPrey2 = 1.50008
fabm.yaml/instances/cladocerans/parameters/fGutOccPrey4 = 4.8261
fabm.yaml/instances/cladocerans/parameters/kDConsMaxZoo = 0.949981
fabm.yaml/instances/cladocerans/parameters/cTmOptZoo = 28.7824
fabm.yaml/instances/cladocerans/parameters/cSigTmZoo = 12.3742
fabm.yaml/instances/cladocerans/parameters/fDAssZoo = 0.301235
fabm.yaml/instances/cladocerans/parameters/fZooDOMW = 0.0515662
fabm.yaml/instances/cladocerans/parameters/kMortZoo = 0.001
fabm.yaml/instances/cladocerans/parameters/kDRespZoo = 0.132183
Original ln likelihood = 110.49561
Copying files for model setup to C:\Users\ADMINI~1\AppData\Local\Temp\gotmopt99csz5oo...
  skipping parsac because it is a directory
  skipping restart.nc because it is a NetCDF file
Evaluating fitness with parameter set [1.2217,1.0077,0.941884,3.28413e-06,0.738787,0.398411,7.40694,0.0629678,0.0416152,
1.03687,1.06267,0.00998945,0.00990978,1.05223,1.04707,0.000159337,2.89222,0.0034557,0.0150304,0.149681,1.51189,0.777001,
1.57426,0.917556,2.08223,1.079,0.000597438,0.000861677,0.670612,1.08654,0.972069,1.05063,1.11325,27.9326,0.00235275,0.00
26888,0.0222718,0.809709,1.95042,0.0184772,1.44527,1.03694,0.039098,1.51618,0.0199577,0.00947401,0.0282785,0.643669,0.64
0192,0.0297333,-0.5,0.0155308,0.0460541,0.0301086,2.1968,25.8694,10.7991,1,7.41933,14.1283,-0.5,0.5,0.247615,1.3709,5.17
096,30.4412,2,35.1659,0.773731,0.483398,23.2962,14.7578,1,11.4078,46.9837,1.56476,1.54922,2.03721,2.48719,1.50008,4.8261
,0.949981,28.7824,12.3742,0.301235,0.0515662,0.001,0.132183].
Starting model run...

-----
GOTM started on 2024/10/04 at 03:50:28
-----

init_gotm

-----
Reading yaml configuration from: gotm.yaml
configuring modules ....
init_airsea_yaml
done
init_stim_yaml
done.

```

parsac created the  
plotbest folder and  
executed the model  
from this folder

```

C:\windows\system32\cmd.exe
Calculating weights for linear interpolation to "wet_totP_calculator_result" observations...done.
Calculating weights for linear interpolation to "wet_totN_calculator_result" observations...done.
Calculating weights for linear interpolation to "wet_chla_calculator_result" observations...done.
Calculating weights for linear interpolation to "cyanobacteria_oChla" observations...done.
Calculating weights for linear interpolation to "diatoms_oChla" observations...done.
Using optimal s.d. for temp = 2.12557.
Using optimal s.d. for abiotic_water_sO2W = 1.27727.
Using optimal s.d. for abiotic_water_sNO3W = 0.206254.
Using optimal s.d. for abiotic_water_sNH4W = 0.0408296.
Using optimal s.d. for wet_totP_calculator_result = 0.0109255.
Using optimal s.d. for wet_totN_calculator_result = 0.161355.
ln Likelihood = 110.496.
Newly calculated ln likelihood = 110.49561. Original value was 110.49561.
temp:
- bias: 0.4878
- mean absolute error = 1.48
- rmse = 2.111
- cor = 0.9743
- s.d. mod = 8.656
- s.d. obs = 7.793
abiotic_water_sO2W:
- bias: 0.7978
- mean absolute error = 1.012
- rmse = 1.268
- cor = 0.9659
- s.d. mod = 3.748
- s.d. obs = 3.448
abiotic_water_sNO3W:
- bias: 0.05684
- mean absolute error = 0.1763
- rmse = 0.2034

```

parsac estimated sd to  
use for weights in  
optimization

State variable  
performance  
calculated by parsac,  
worth saving for each  
calibration iteration

Users can also execute second, third and so on best performing model parameterizations by specifying model rank: “parsac calibration plotbest.xmlfile -r RANK”.

### *parsac plot*

The plot function creates a “dotty”-plot that shows the relative ln likelihood against the model simulation’s parameter values in subplots for each parameter. So, each dot in a subplot represents a relative model performance (that is equal across subplots and parameters) against a parameter value within the parameter values ranges specified in the xml file.

The best performing parameter range is indicated by parsac with vertical dashed lines.

In large database files, it is not convenient to plot all model simulations with plot function. Users can specify a lower boundary for relative ln likelihood (always < 0) to select subset of model simulations with “-r RANGE”: “parsac calibration plot xmlfile -r RANGE”.

A calibration “signal” can be interpreted as a parameter range where many of the dots fall within close the relative ln likelihood of 0. The strength of this signal can vary, and it is up to the modeler to interpret if the parameter ranges should be expanded, narrowed, remain the same or shifted in the next calibration iteration.

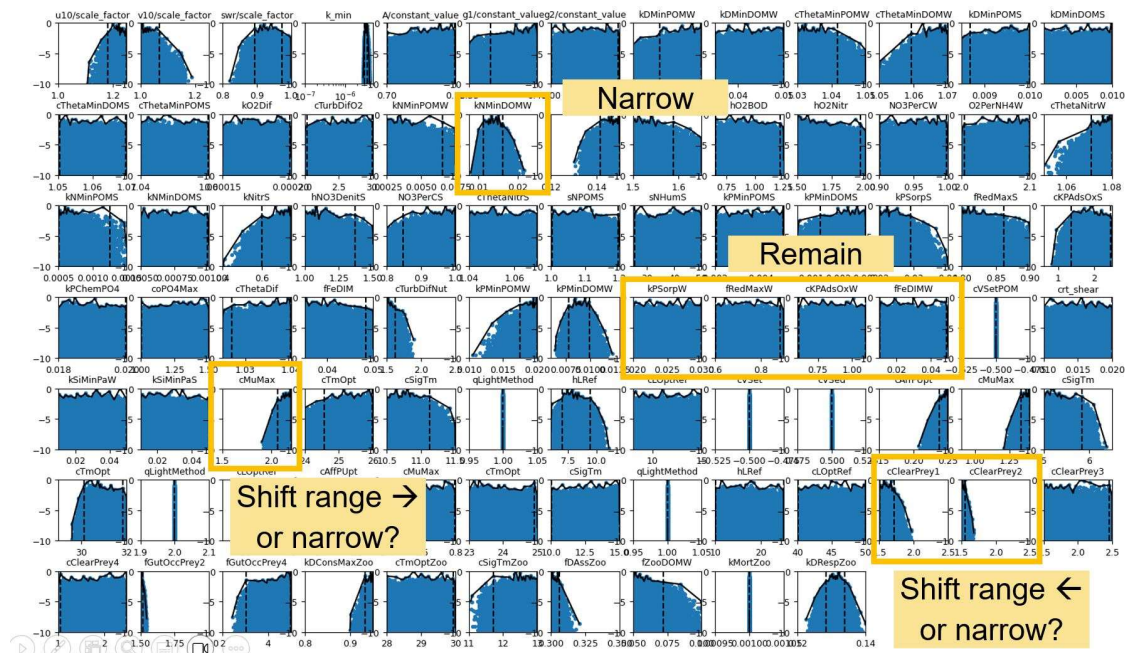


Figure 1. Dot-plot from one calibration iteration for Shahe Reservoir with approx. 1 million model simulations in the database file. The range function has been used, so only the relative ln likelihood from 0 to 10 is plotted. Potential interpretations of the calibration signal of some parameters are included in orange.

The number of parameters with a calibration signal and the strength of the signal depends on the number of model simulations in the database file.

You should evaluate how the calibration signal and parameter range performances compare with the overall model behavior and simulated ecosystem dynamics. For this, we recommend to plot time series plots of state variables and observations, as well as other model evaluation plots, and evaluate if the calibration signal and parsac recommendations would increase the GOTM-WETS ability to reproduce the aquatic ecosystem dynamics and properties for the right reasons (i.e. parameter values are within technical and theoretical boundaries). parsac “recommendations” is model-agnostic and so all users should understand if shifts in parameter value ranges is within model theory and is realistic.

#### QWET plot diagnostics

QWET diagnostics plot (under the tab “obs” in QWET) is one such visualization tool that plots many common model evaluation visualizations.

### Calibration strategy and iterations

To facilitate the calibration process, the calibration procedure will typically be divided into a series of steps, where each step adds additional parameters (and their ranges). Each step will target a selection of parameters that will have strong influence on the temporal and vertical dynamics of specific state variables. The aim is to achieve a strong signal in the objective function based on calibration of only a subset of model parameters, and thereby facilitate and enhance the ability to narrow the ranges of individual parameters following each calibration iteration. A somewhat generic approach would be to target (in this order): physical dynamics, e.g. temperature dynamics (and sediment resuspension if relevant, e.g. typically for shallow lakes) -> oxygen dynamics -> nutrient dynamics -> phytoplankton dynamics (and higher trophic levels). As WET is a complex ecosystem model, many parameters interact and so it is recommended to go through a calibration step approach and spend most of the time calibrating all selected parameters against all included observations.

You may find yourself needing to go back to a previous calibration iteration. For instance, if you learn something new about your case study and/or have constrained some parameter value ranges too much. Therefore, we recommend you save model folders from plotbest, model performance metrics, doty-plots, xml files and optionally database files for each iteration, so you have a library and documentation of the calibration process.

### Bat files (optional)

It is convenient to create a few bat files that will help you easily run ParSac. These bat files could be placed in the folder of you model set up. See the example xml file (which can be used and modified)



for how to configure the calibration.

#### PARSAC CALIBRATION RUN

Example of content of bat for running “parsac calibration run” (in one line):

C:\Python3\Scripts\parsac calibration run

C:\Data\lake\_models\lake\_example\parsac\config\_parsac.xml

#### PARSAC CALIBRATION PLOT

Example of content of bat for running “parsac calibration plot” (in one line):

C:\Python3\Scripts\parsac calibration plot

C:\Data\lake\_models\lake\_example\parsac\config\_parsac.xml

#### PARSAC CALIBRATION PLOT -r

*(Plot results with a zoom to higher/better likelihoods (dotty plots, i.e. parameter values vs objective function))*

Example of content of bat for running “parsac calibration plot -r” (in one line):

C:\Python3\Scripts\parsac calibration plot -r -200

C:\Data\lake\_models\lake\_example\parsac\config\_parsac.xml

#### PARSAC CALIBRATION PLOT -u (Plot results that will be updated on-the-fly)

Example of content of bat for running “parsac calibration plot -u” (in one line):

C:\Python3\Scripts\parsac calibration plot -u

C:\Data\lake\_models\lake\_example\parsac\config\_parsac.xml

#### PARSAC CALIBRATION PLOTBEST (Plot results from the currently best performing simulation)

Example of content of bat for running “parsac calibration plot plotbest” (in one line):

C:\Python3\Scripts\parsac calibration plotbest

C:\Data\lake\_models\lake\_example\parsac\config\_parsac.xml

#### PARSAC CALIBRATION PLOTBEST -r (Plot results from the currently a specific rank of the best performing simulations)

Example of content of bat for running “parsac calibration plot plotbest” (in one line):

C:\Python3\Scripts\parsac calibration plotbest -r 2

C:\Data\lake\_models\lake\_example\parsac\config\_parsac.xml